

Human Computer Interaction (HCI, HCC)

AN INTRODUCTION



Human Computer Interaction

Why are we here?

- It may seem trite, but user interfaces matter: For efficiency, for convenience, for accuracy, for success, even for life and death
- The Vincennes, and Aegis RADAR system

Human Computer Interaction

Why are we here?

- It may seem trite, but user interfaces matter: For efficiency, for convenience, for accuracy, for success, even for life and death
 - The Vincennes, and Aegis RADAR system
- People time is more expensive than computer time (normally)
- Everyone has a story of a bad interface, and the trouble it caused
- A well-designed interface makes for a good system, and a happy (and efficient) user



(This is a bad interface)

Human Computer Interaction

Sub-discipline of the multi-discipline Human Factors and Ergonomics

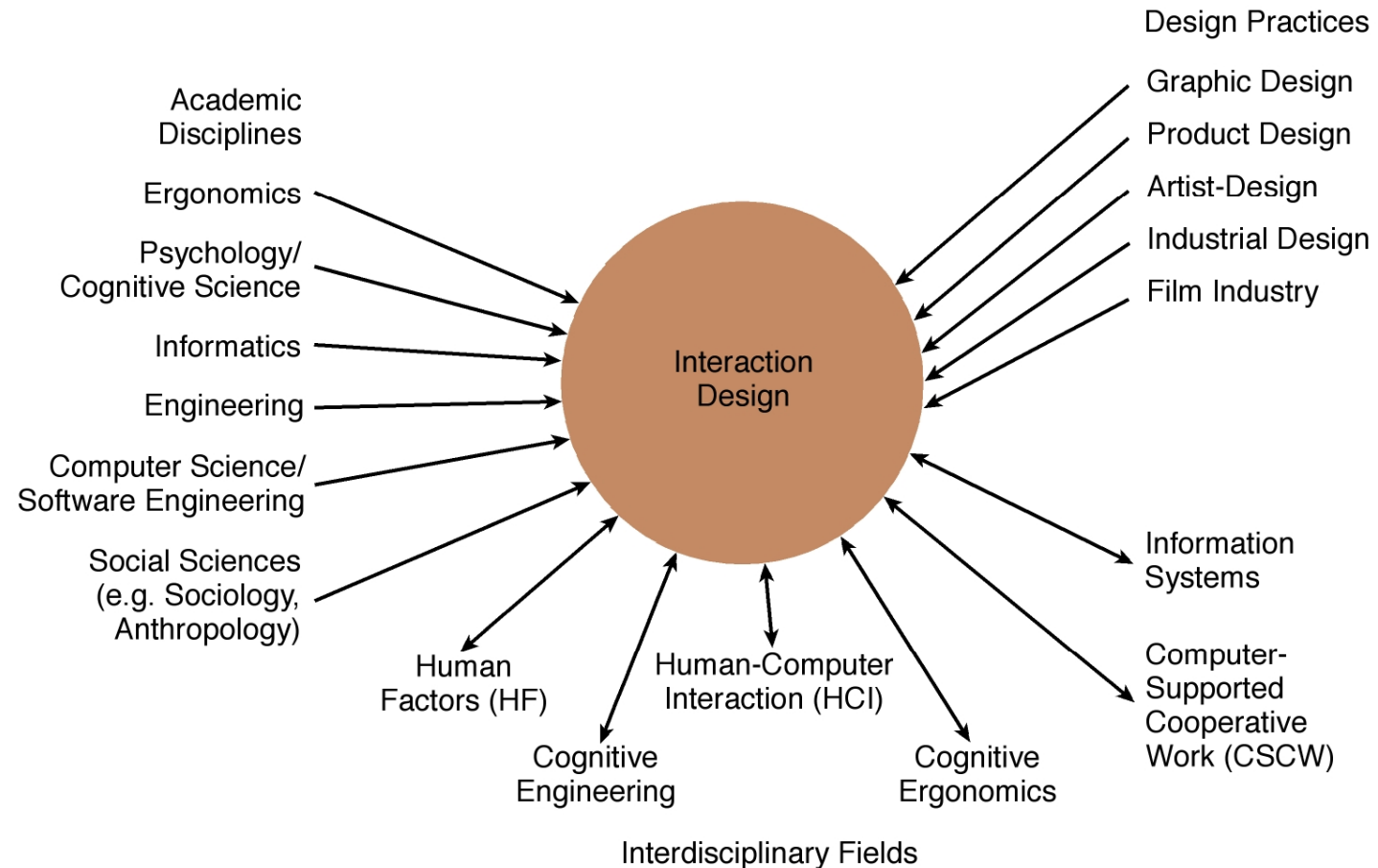
- Psychology (many types)
- Design
- Engineering
- Social sciences

Both deal with the design of everyday (and not-so-everyday) things

Involve questions such as, but not limited to:

- Is it easy to use?
- Is it easy to understand?
- Is it easy to discover?
- Is it comfortable to use?
- Is it sensible to use?

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Each word is important

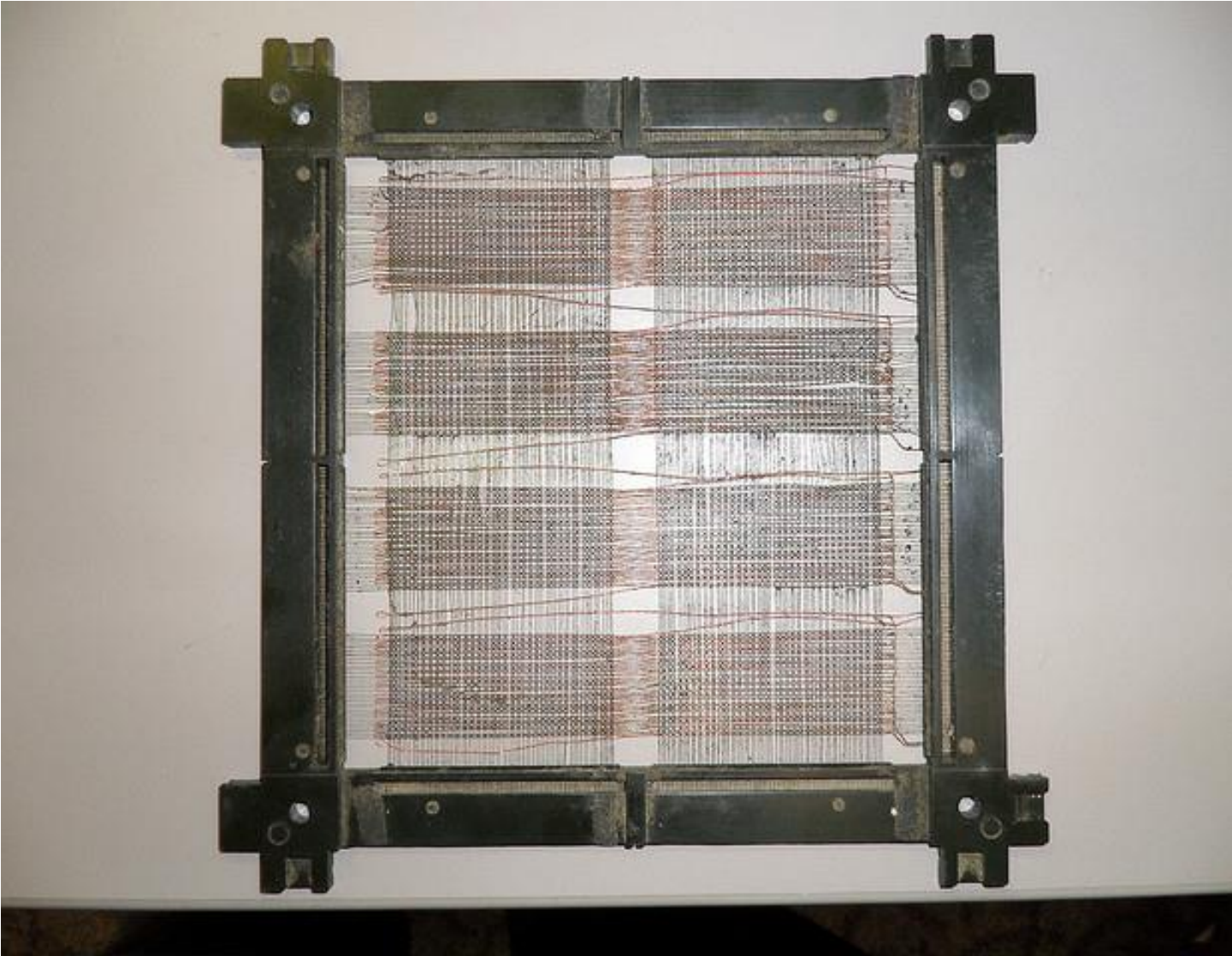
In order to understand the discipline, we must understand:

- The Human
- The Computer
- The Interaction

These three components come together (interact!) to form what will hopefully be the User experience

None of these components can be left out, or ignored, they are all required for the development of a successful interface

A well-developed and designed interface is of singular importance, because it is the *sole gateway* to any system



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For many years, this was not a consideration

- In other words, what it was, was nothing

Very early on, machines had to be hardwired

Human Computer Interaction

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Very early on, machines had to be hardwired

Later, commands were entered one line at a time

- Attempts were made to make this easier, but that was difficult due to technological limitations
- Some intuitiveness was included, but not much
- Still early, and in large part still the domain of hobbyists and scientists
- Cryptic, undecipherable code
 - “The Software Crisis”
 - Can have long-lasting impacts, as seen in the Y2K problem

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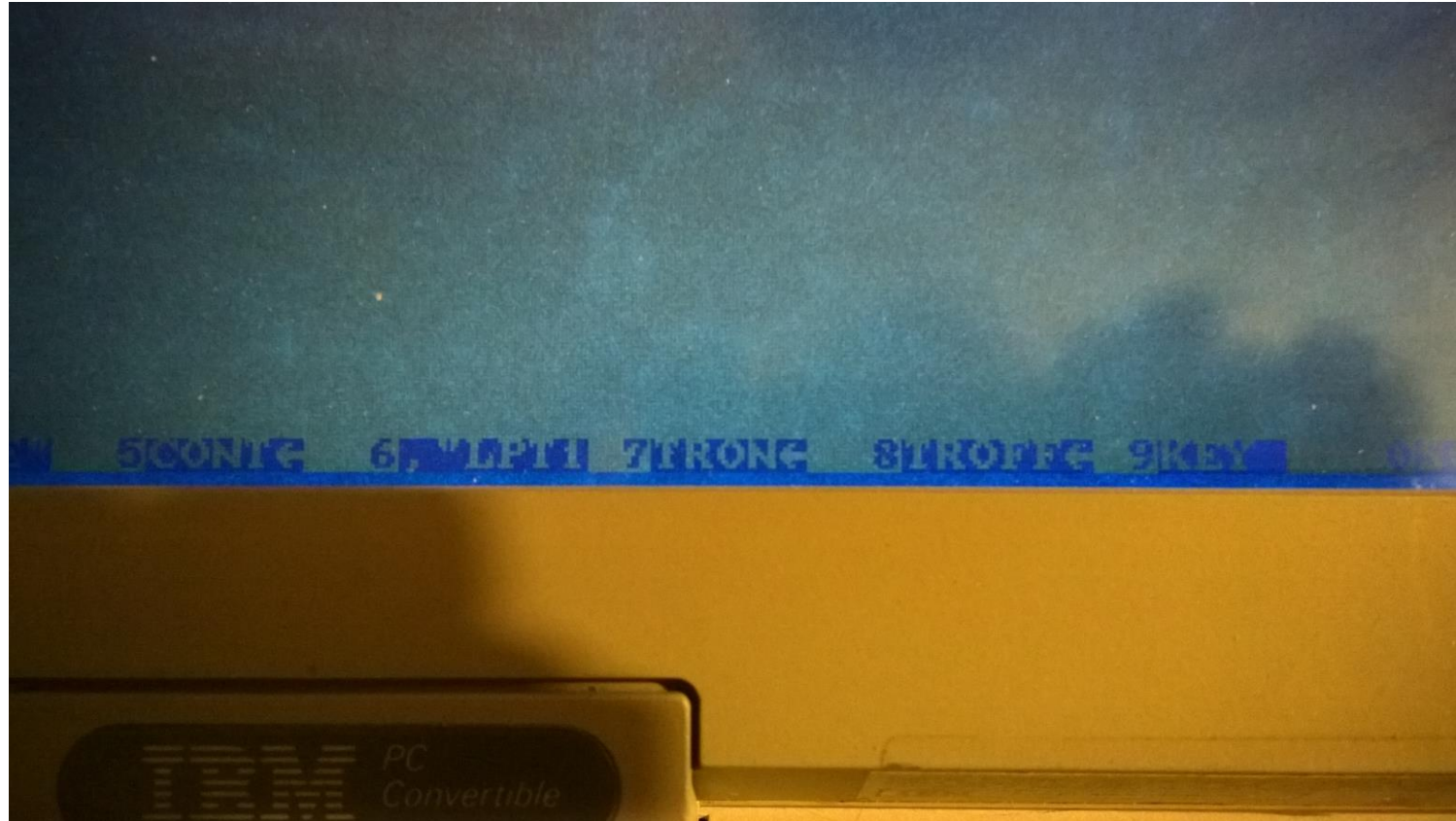
Starting MS-DOS...

C:\>_

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```
SELECT COMMANDS OPTION AS FOLLOWS:  
  
OPTION #1 : GRAPHIC COMMANDS BUT NO  
            'LET' OR 'REM' COMMANDS  
  
OPTION #2 : 'LET' & 'REM' COMMANDS BUT  
            NO GRAPHICS  
  
WHICH OPTION # DO YOU WANT ?1  
COPYRIGHT 1977 BY APPLE COMPUTER INC.  
  
MEMORY SIZE? 25693  
_14940 BYTES FREE  
J
```

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However, this time period also saw expansion of the field of ergonomics, HCI, and the disciplines involved

Originally used primarily for word processors and spreadsheets

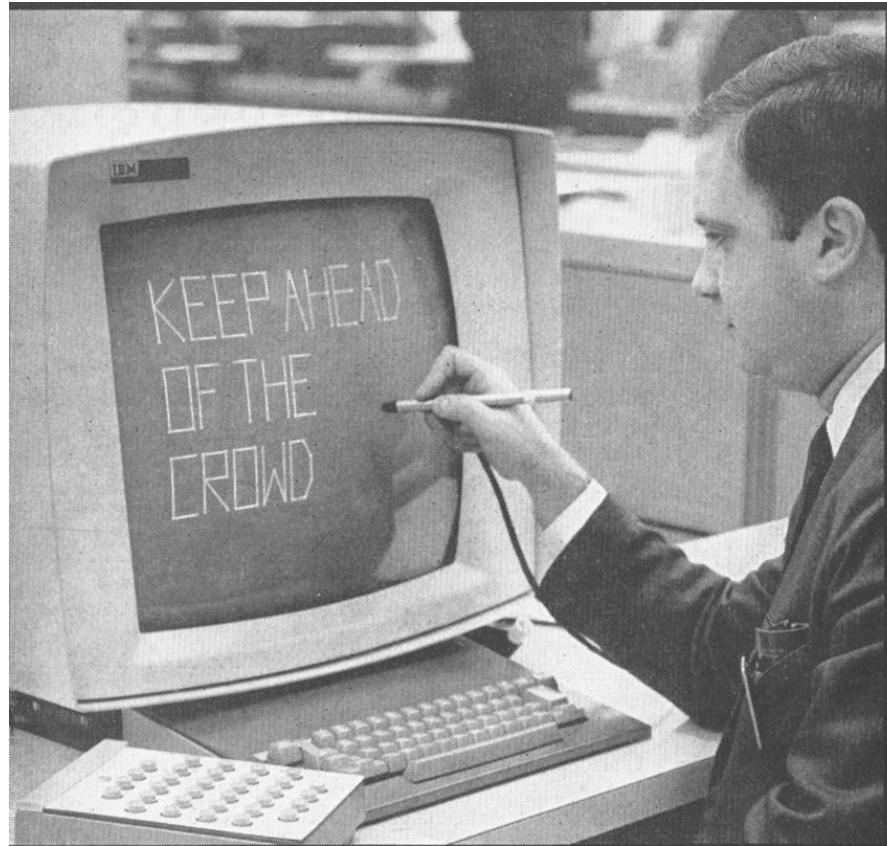
- Still very cumbersome to use
- Often required many-key combinations to perform tasks
- Keyboard overlays were standard
- This was all a result of, but also evolution of, the interface

A slow movement away from technical manuals, and towards user manuals, was also budding

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C11 (L) TOTAL				C1
				25
	A	B	C	D
1	ITEM	NO.	UNIT	COST
2	---	---	---	---
3	MUCK RAKE	43	12.95	556.85
4	BUZZ CUT	15	6.75	101.25
5	TOE TONER	250	49.95	12487.50
6	EYE SNUFF	2	4.95	9.90
7				
8			SUBTOTAL	13155.50
9			9.75% TAX	1282.66
10				
11			TOTAL	14438.16
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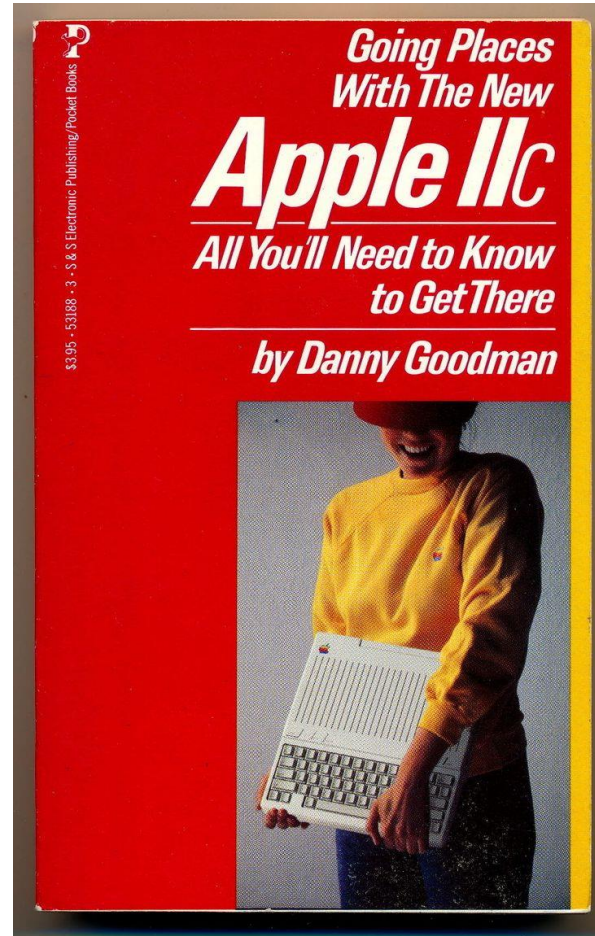
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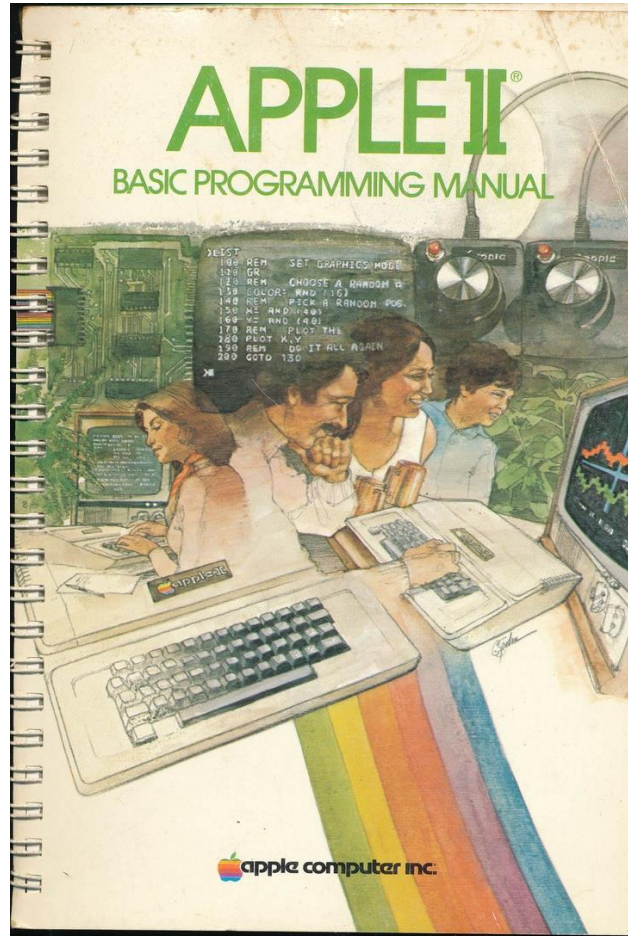
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Most importantly, system and software design and development began to focus on the experience, as opposed to just the hardware and software

- Usability
- Maintenance / Support
- Graphical interface
- All of which end at the user
 - Understanding them
 - Listening to them
 - Designing for them

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Usability is really the ultimate focus, the ultimate goal

But what does it mean?

The term is a nebulous, living thing

- Frequently changing
- Frequently expanding
- Frequently being added to

Usability doesn't mean simply 'able to be used.'

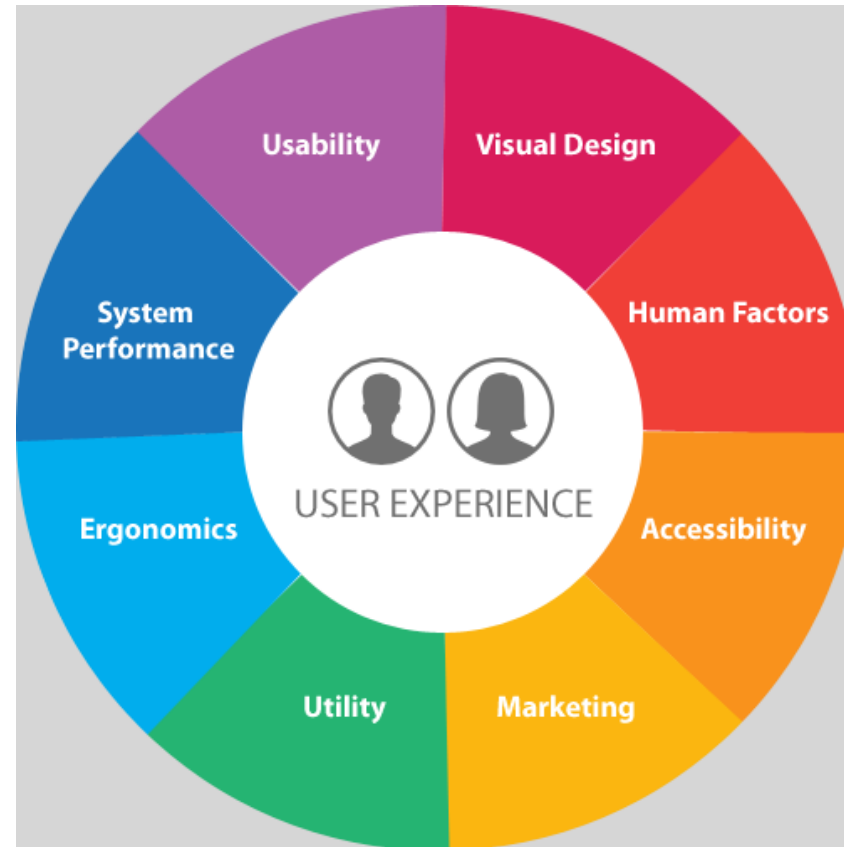
- Distinction between UI/UX

HCI's multidisciplinary nature can lead to complexities in approach and definition

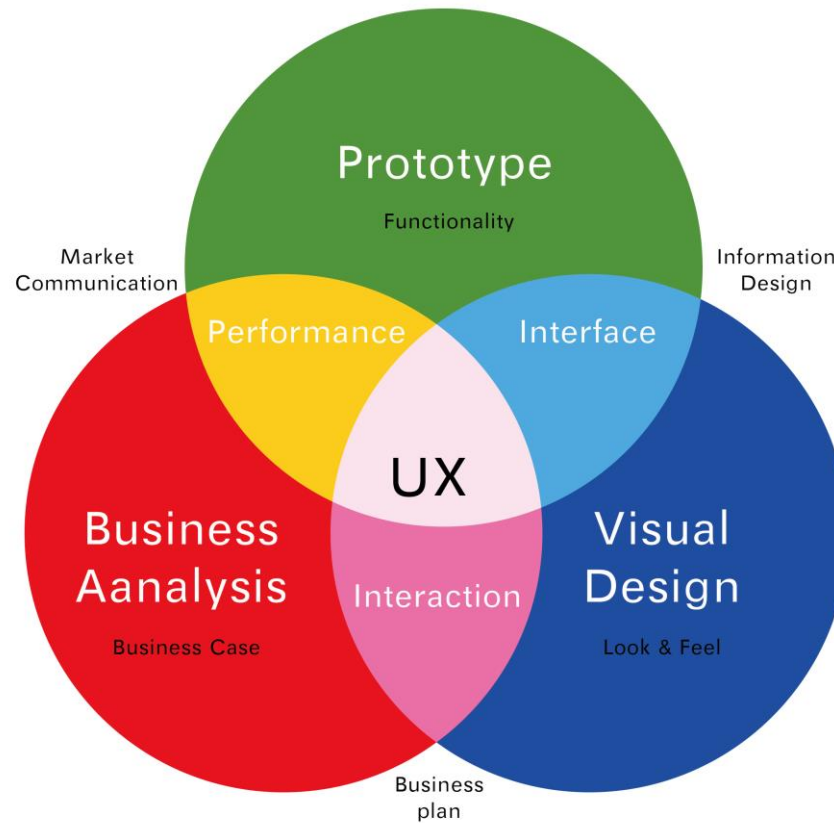
HCI (Human Factors, really) is also one of the most co-opted and diluted disciplines there is

- However that also gives it its strength

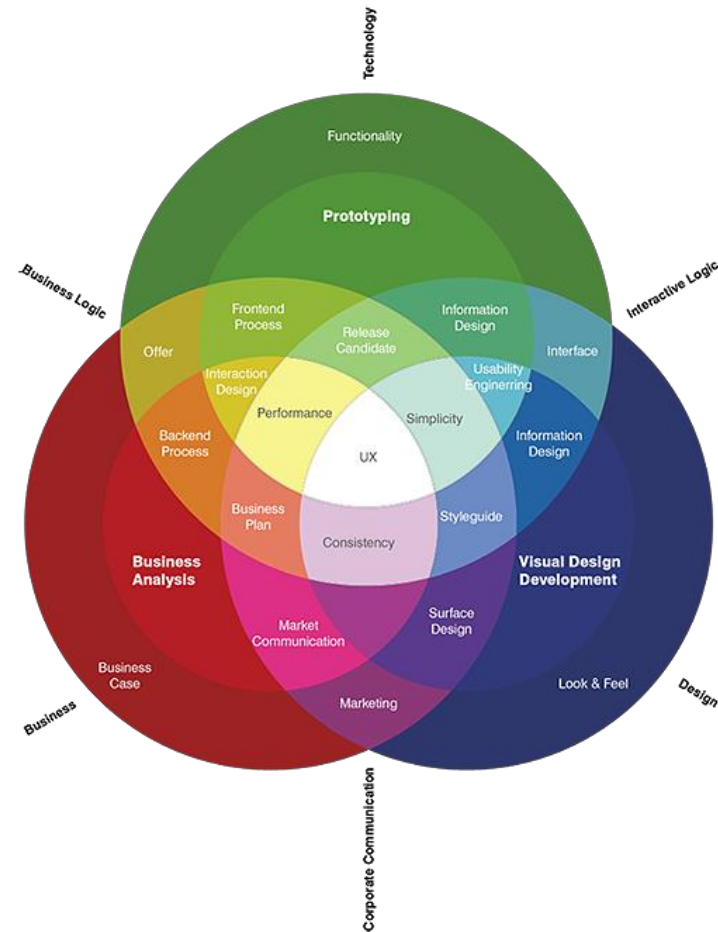
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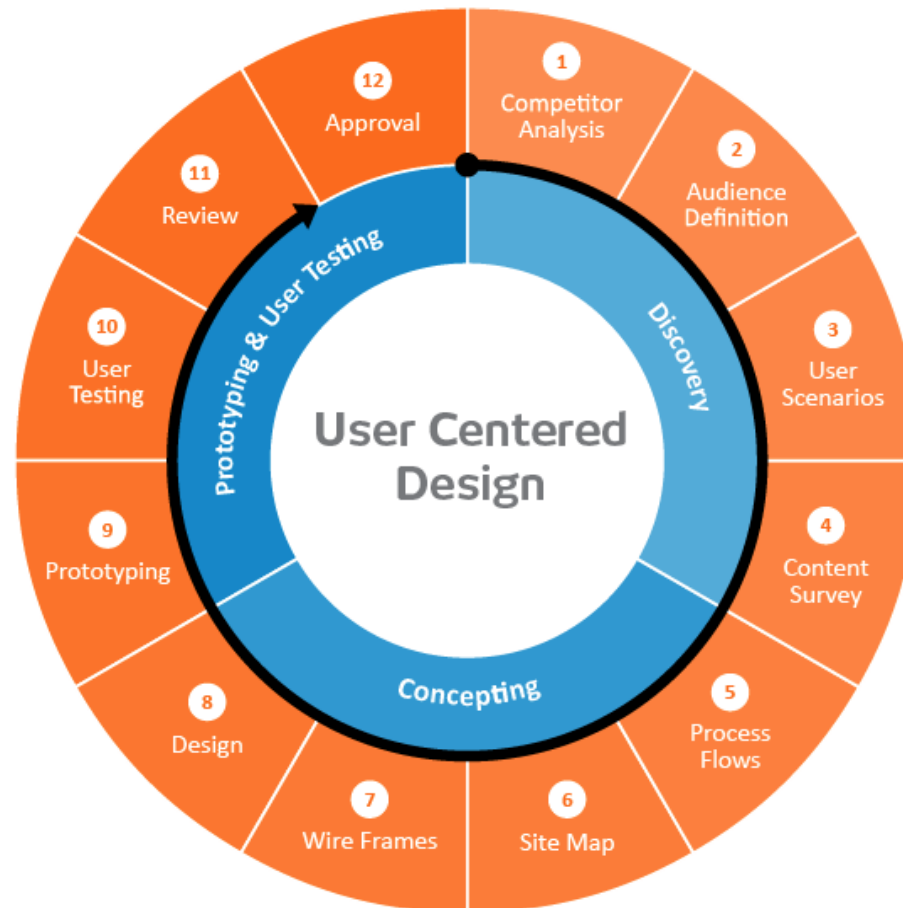
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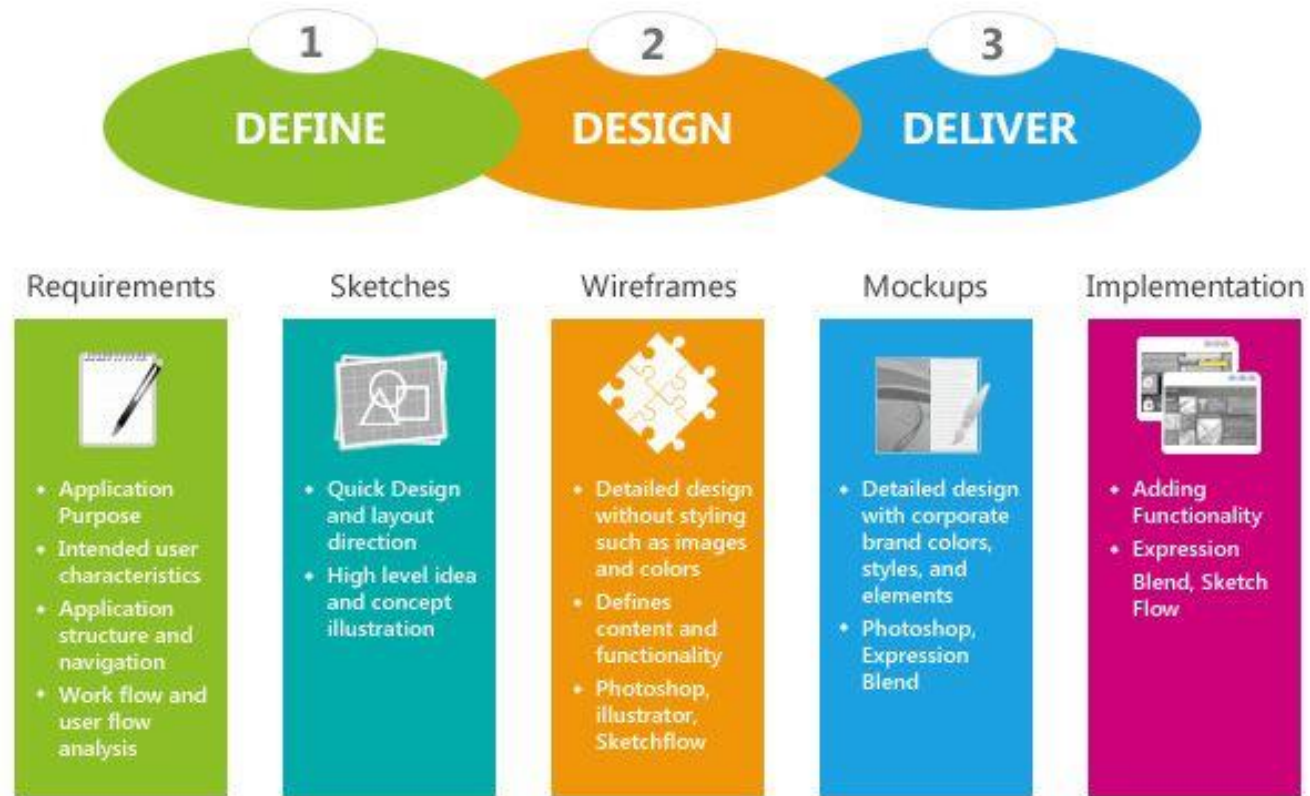
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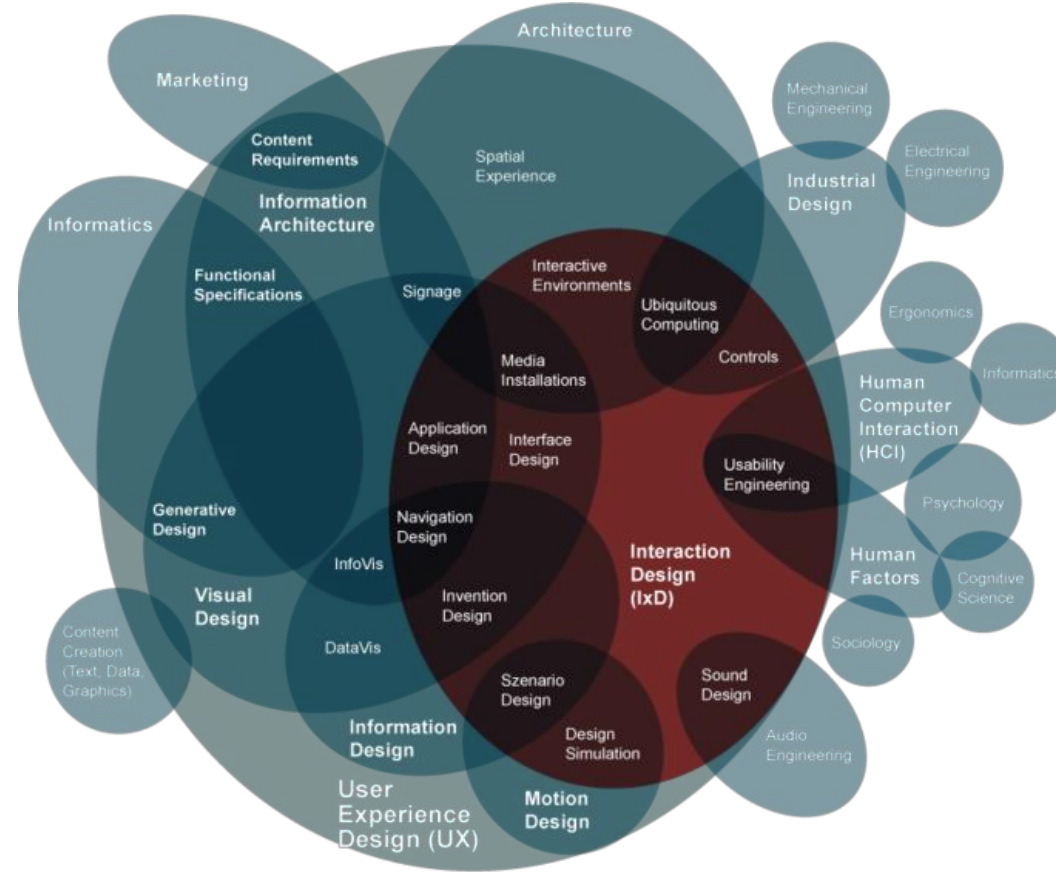
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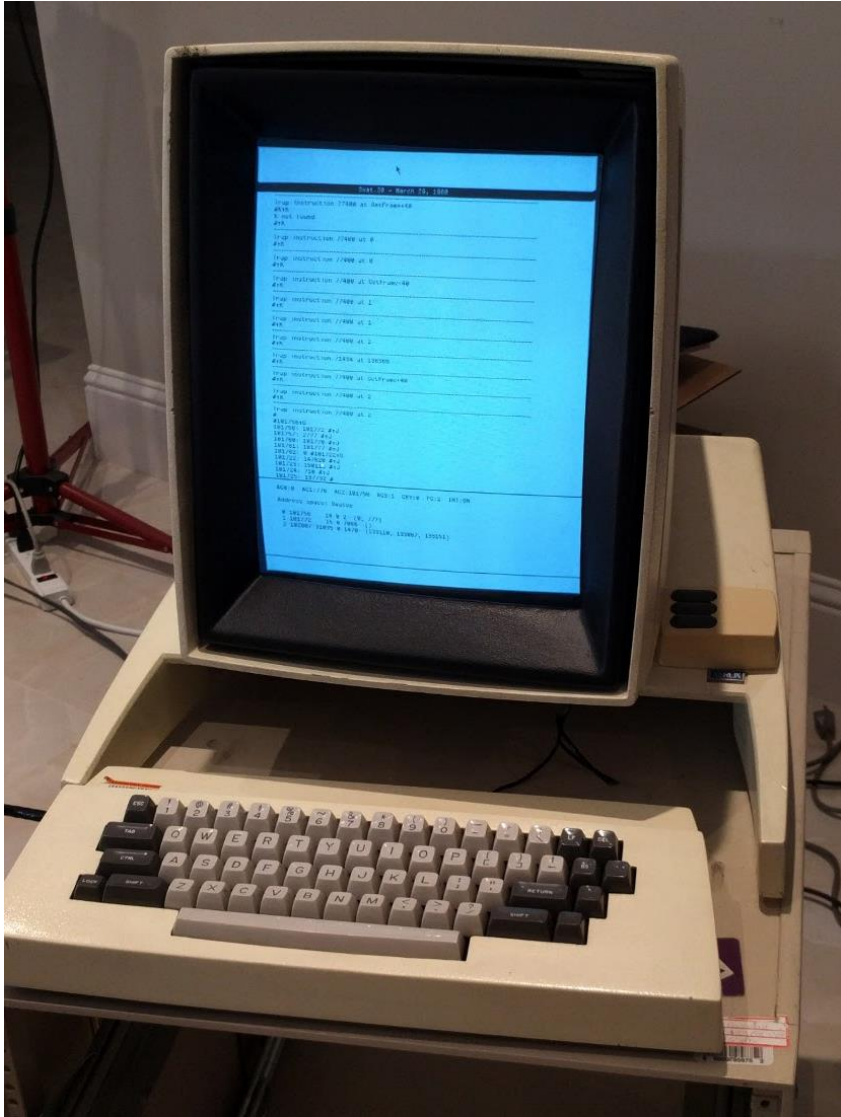
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www.kickerstudio.com/blog/2008/12/the-disciplines-of-user-experience

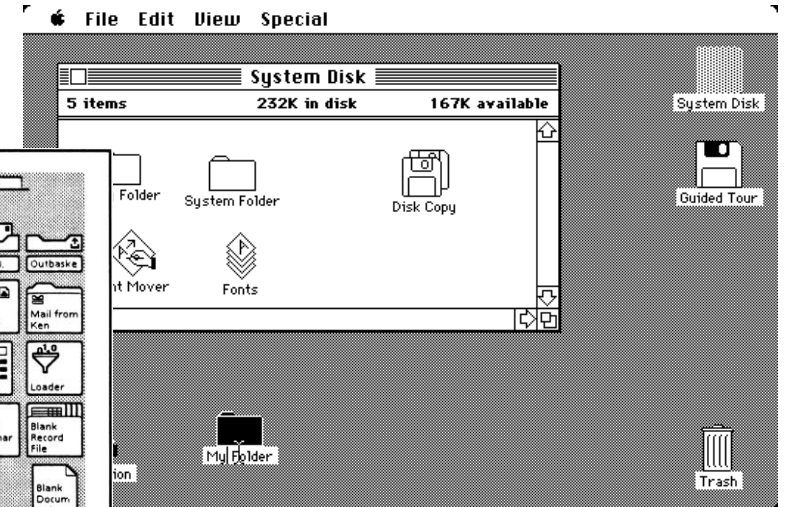
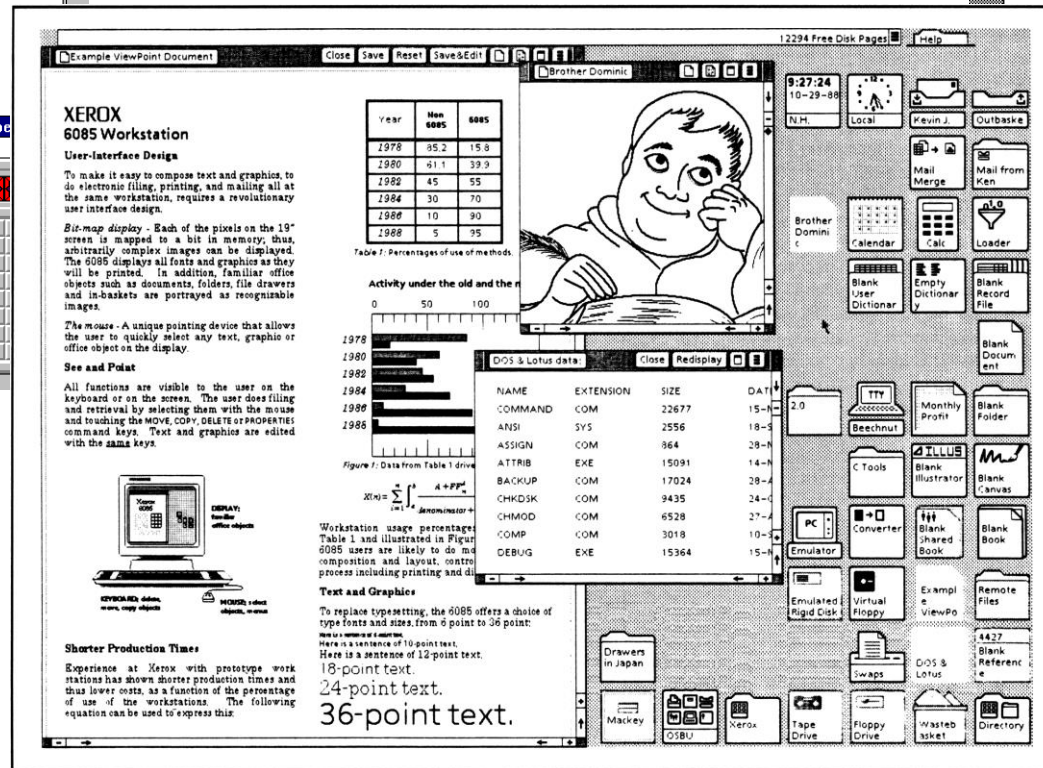
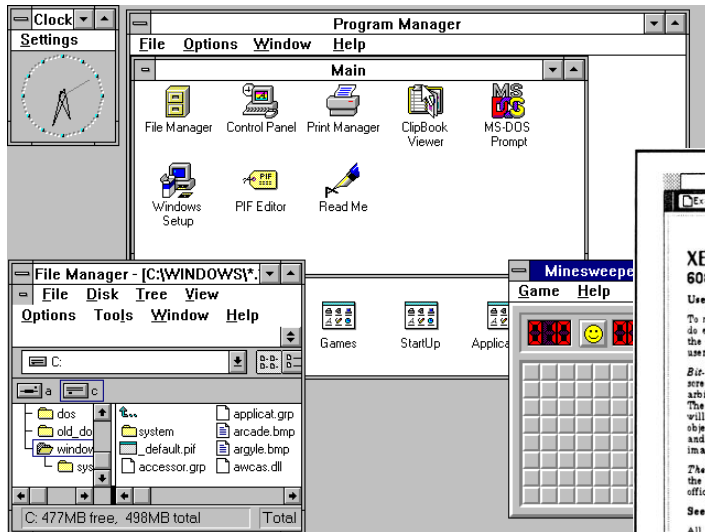


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Eventually, there was a sea change

- Xerox Star
- Apple Macintosh
- Microsoft Windows

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Provided brand new methods of interaction

- WIMP

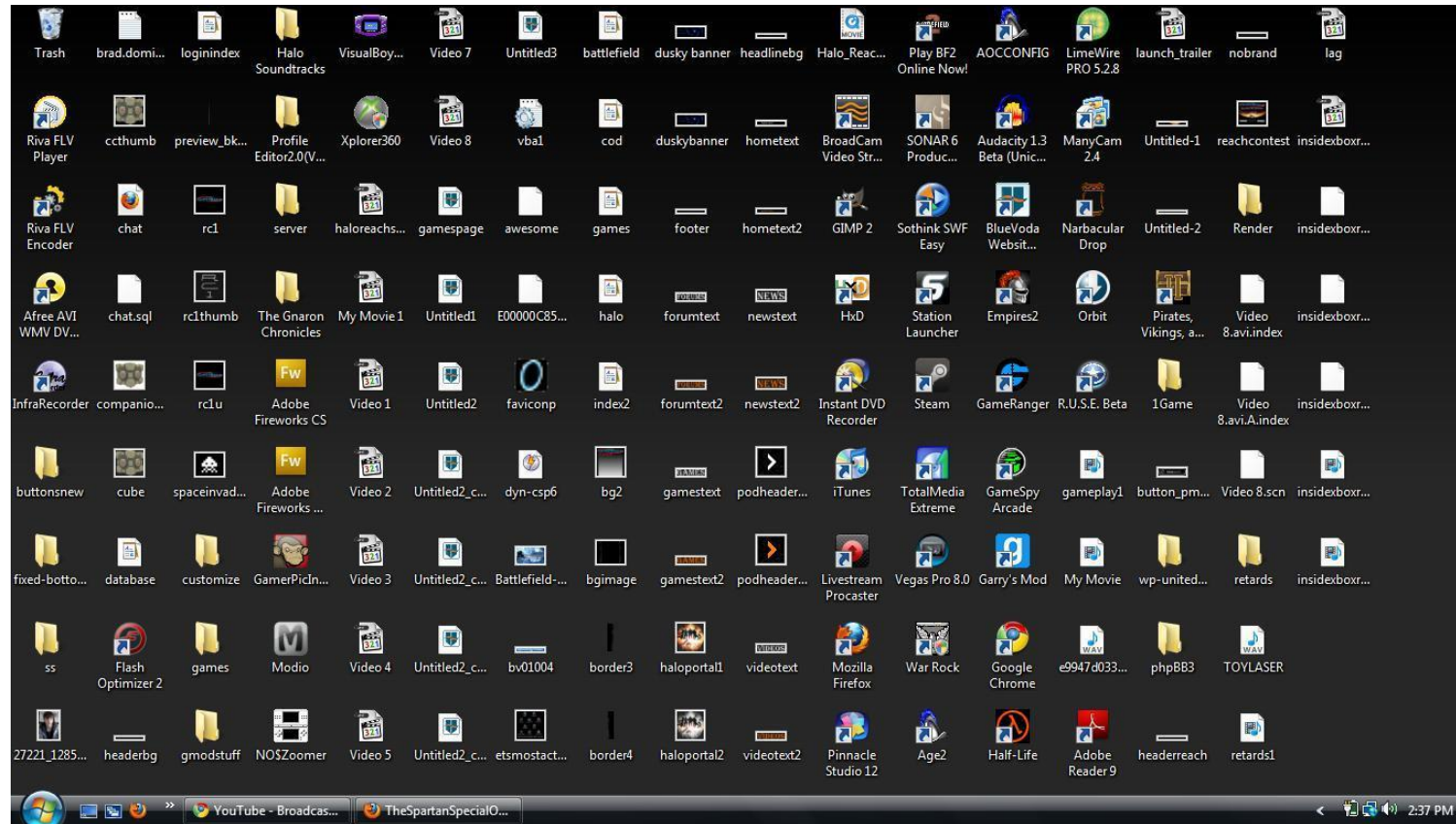
New interaction types

- Instructing
- Conversing
- Manipulating
- Exploring

This led to an explosion in HCI, in all areas

- New interaction methods
- New conceptual models
 - Metaphors
 - Affordances
- Increased accessibility, and also increased accessibility
- This also led to a host of new problems

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Standards and guidelines have been developed and incorporated into interface design for decades

However, two researchers have contributed sets of rules that have been tested, verified, and incorporated more than any other

Jakob Nielsen

- 10 Usability Heuristics for Interface Design

Ben Schneiderman

- The Eight Golden Rules of Interface Design

Each of these are considered canon in the design field

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The 10 Usability Heuristics for User Interface Design

- Visibility of system status
- Match between system and the real world
- User control and freedom
- Consistency and standards
- Error prevention
- Recognition rather than recall
- Flexibility and efficiency of use
- Aesthetic and minimalist design
- Help users recognize, diagnose, and recover from errors
- Help and documentation

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The Eight Golden Rules of Interface Design

- Strive for consistency
- Cater to universal usability
- Offer informative feedback
- Design dialogs to yield closure
- Prevent errors
- Permit easy reversal of actions
- Support an internal locus of control
- Reduce short term memory load

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Usability goals

- Effectiveness
- Efficiency
- Safety
- Utility
- Learnability
- Memorability

User experience goals

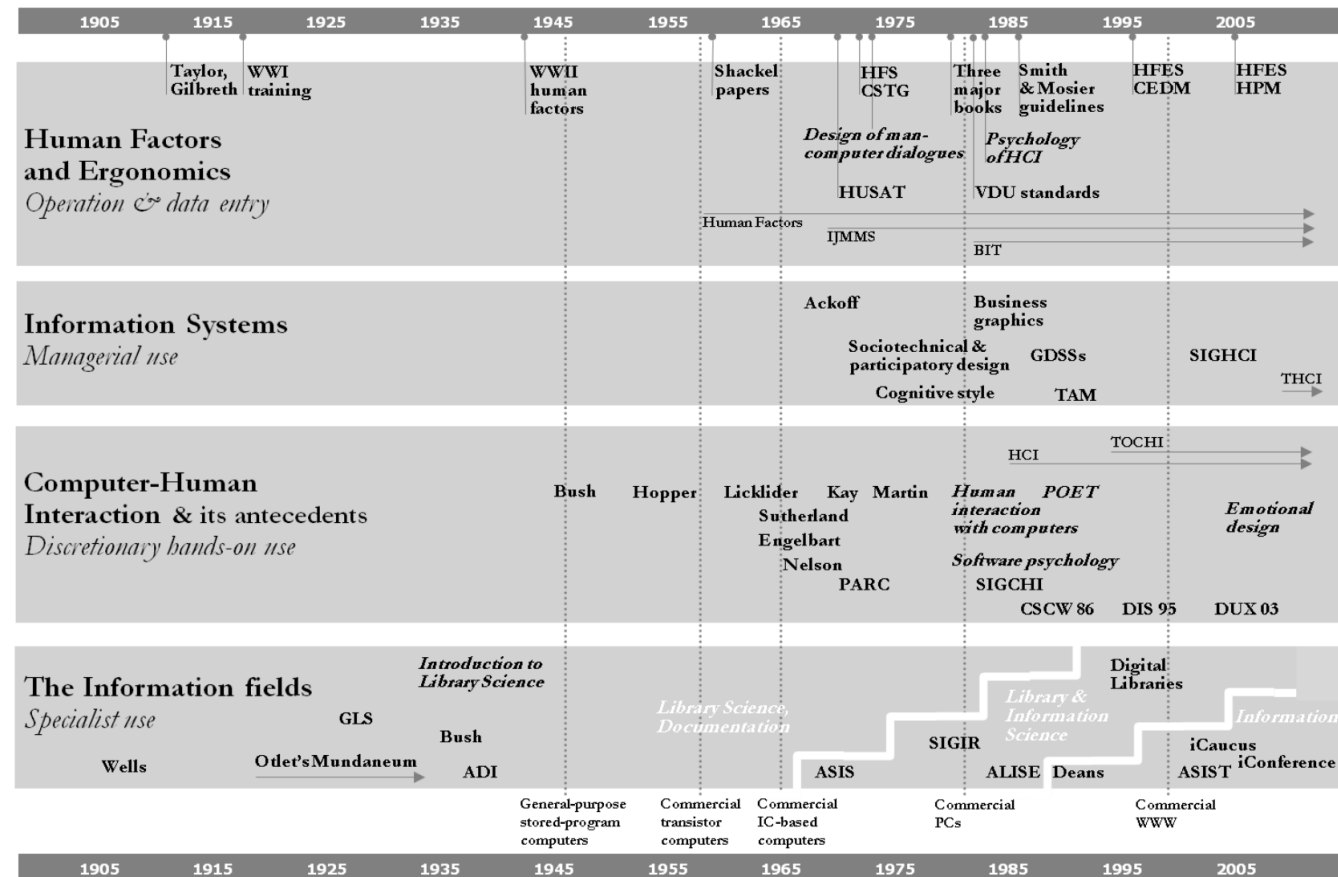
- Desirable aspects
- Undesirable aspects

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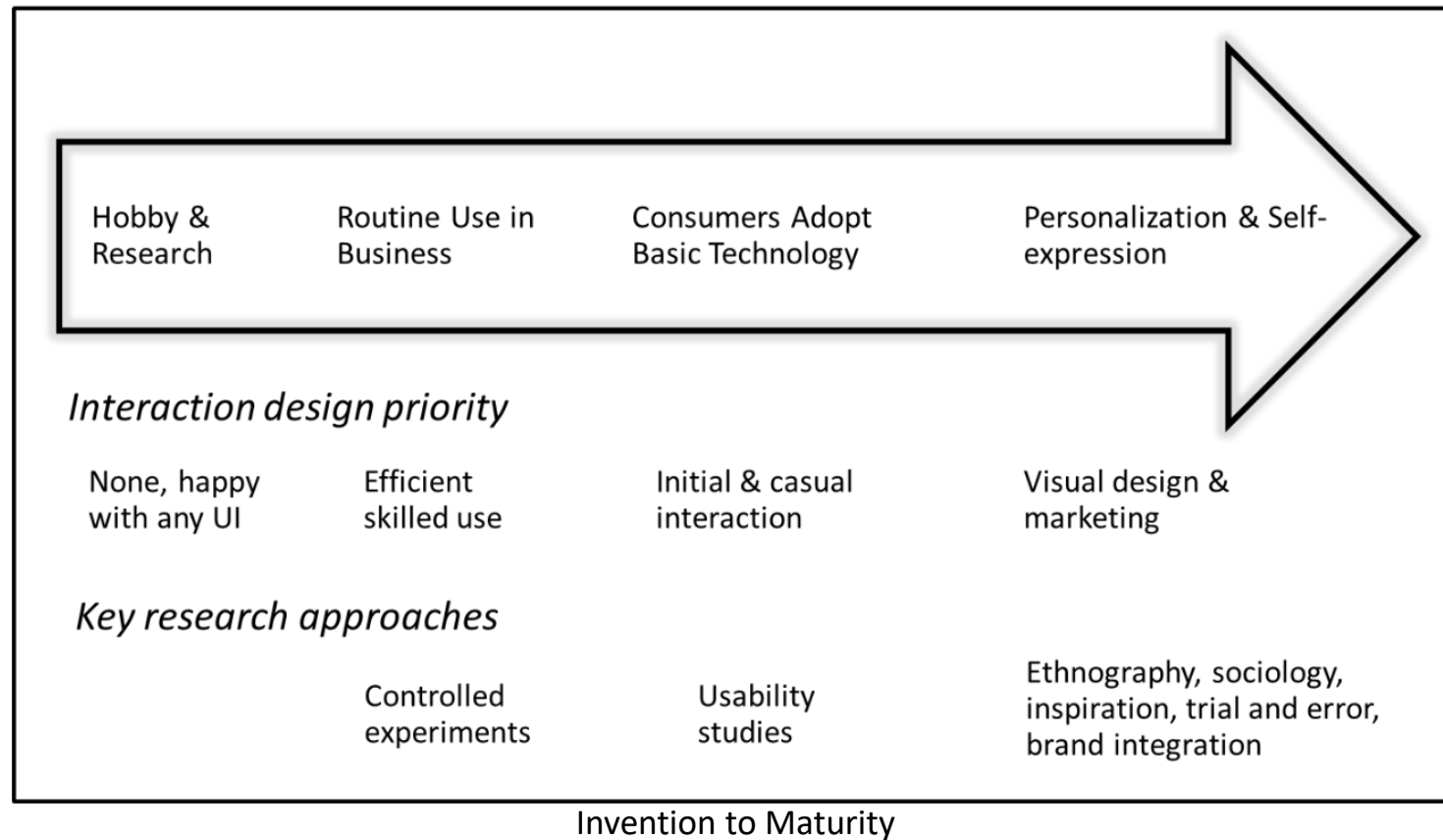
Eventually, there was movement beyond the PC

- How do we address the nearly infinite amount of objects we can store?
- Internet (Command to graphical, Lynx to browsers)
- Communication / Collaboration
 - LOL U WOT M8
- Evolution and place of the computing environment
 - Background devices
 - Used by everyone (and why?)
 - Significant functionality in individual devices
 - Embedded systems
- With this evolution, the discipline evolves as well

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With the increase in user-generated content, and the evolution of 'Web 2.0,' there has become a drastic increase in the need to access, manage, display, and manipulate the vast amounts of data available to us

Ackoff addressed the handling of computer-generated information back in 1967

- Five assumptions
 - Decision makers lack relevant information
 - Decision makers need information they want
 - Needed information leads to better decisions
 - Better communication among decision makers leads to better decisions
 - How the IS works is less important than how to use it

Human Computer Interaction

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How did we do it before?

- File cabinets
- Ledgers
- Index cards / card catalog

Operating computers was a specialized affair

- Programmers
- Keypunchers
- Computer operator

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1970s

- User manuals
- Testing labs
- Prototypes
- Goal-oriented design

1980s

- Research began on VDTs
- General design guidelines were first published
- ANSI standards were first developed

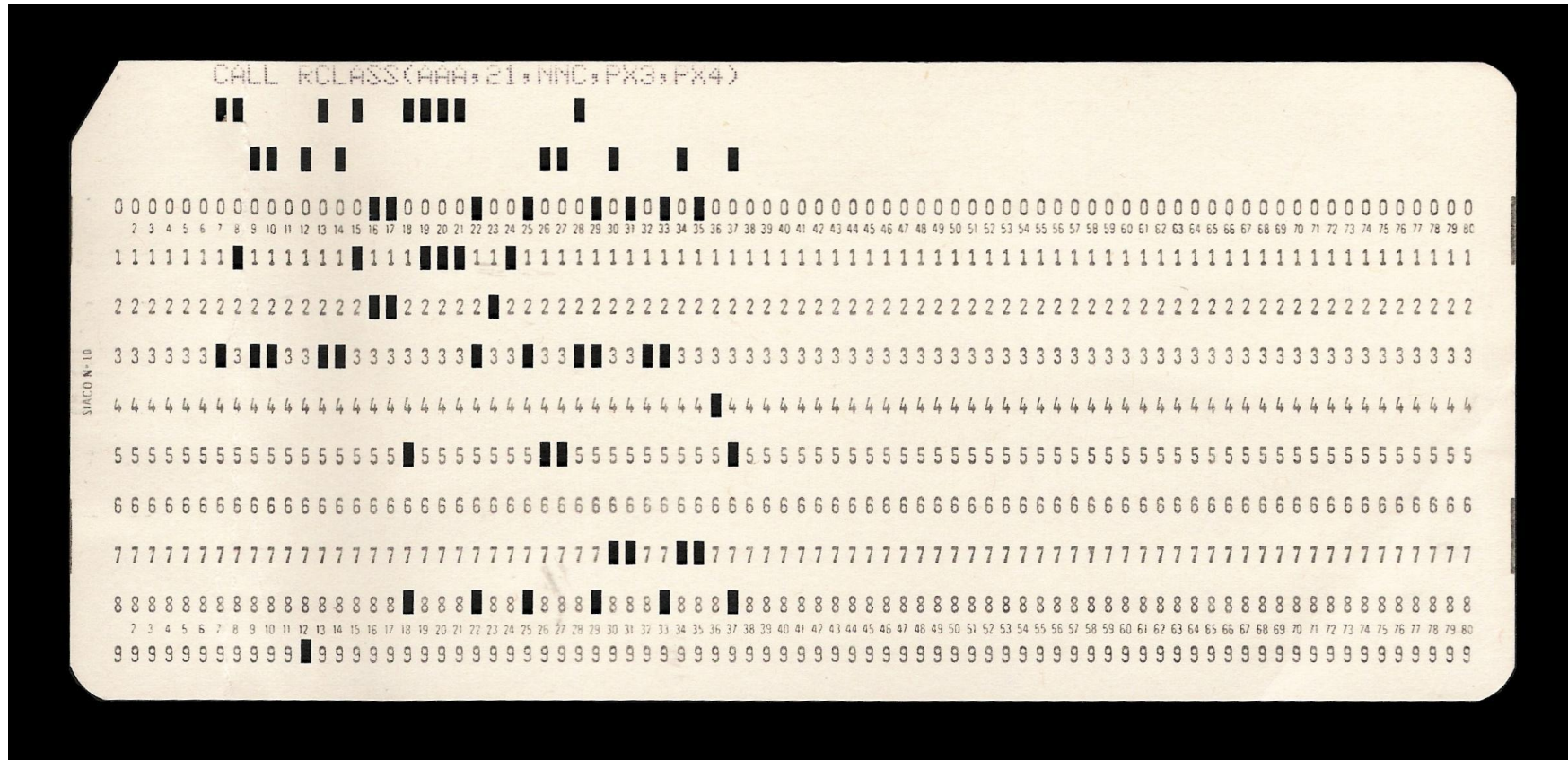
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With the increase in user-generated content, and the evolution of 'Web 2.0,' there has become a drastic increase in the need to access, manage, display, and manipulate the vast amounts of data available to us

As storage capacity increased, data increased, and our ability to use that data increased

- Punch cards

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- Punch cards
- Magnetic media

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- Optical media

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- Cloud

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- Magnetic media
- Optical media
- Cloud
- How do we interact with all this information?

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Now, HCI applies to many types of systems, interactions, and experiences

- Medical information systems / equipment
- Accessibility (Blind, deaf, infirmed, incapacitated)
- Gaming
- Large information environments (cockpits, power plants)
- Social media
- Smart TVs
- Portable devices
- Education
- Military
- Hard sciences
- Business / Manufacturing

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What about the future?

- Artificial intelligence
- Virtual reality
- Ubiquitous systems
- Autonomous vehicles (planes too? Maybe boats?)
- Cybernetics

How will we interact with these?

How will we design these systems so we *can* interact with them?

Ultimately, who will be in control? Can we maintain control? Do we want to?